

UNIVERSITÀ
DEGLI STUDI
DI BERGAMO

Dipartimento
di Scienze Aziendali

Responsibility Accounting Systems

Excercises

Ex 1

Provide the missing data in the following table for a distributor of martial arts products:

	Division		
	Alpha	Bravo	Charlie
Sales	$?	$11,500,000	$?
Net operating income	$?	$ 920,000	$210,000
Average operating assets	$800,000	$?	$?
Margin	4%	?	7%
Turnover	5	?	?
Return on investment (ROI)	?	20%	14%

Ex 1

	Division		
	Alpha	*Bravo*	*Charlie*
Sales (a)	$4,000,000	$11,500,000 *	$3,000,000
Net operating income (b).....	$160,000	$920,000 *	$210,000
Average operating assets (c).....	$800,000 *	$4,600,000	$1,500,000
Margin (b) ÷ (a)	4%*	8%	7%*
Turnover (a) ÷ (c).....	5*	2.5	2
Return on investment (ROI).	20%	20%*	14%*

Note that Divisions Alpha and Bravo apparently have different strategies to obtain the same 20% return. Division Alpha has a low margin and a high turnover, whereas Division Bravo has just the opposite.

*Given.

Ex 2

Tan Corporation of Japan has two regional divisions with headquarters in Osaka and Yokohama. Selected data on the two divisions follow:

	Division	
	Osaka	Yokohama
Sales	$3,000,000	$9,000,000
Net operating income	$210,000	$720,000
Average operating assets	$1,000,000	$4,000,000

Required:

1. For each division, compute the margin, turnover, and return on investment (ROI).
2. Compute the residual income for each division assuming the company's minimum required rate of return is 15%.

Ex 2

1. ROI computations:

$$\text{ROI} = \frac{\text{Net operating income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Average operating assets}}$$

Osaka Division:

$$\text{ROI} = \frac{\$210{,}000}{\$3{,}000{,}000} \times \frac{\$3{,}000{,}000}{\$1{,}000{,}000} = 7\% \times 3 = 21\%$$

Yokohama Division:

$$\text{ROI} = \frac{\$720{,}000}{\$9{,}000{,}000} \times \frac{\$9{,}000{,}000}{\$4{,}000{,}000} = 8\% \times 2.25 = 18\%$$

Ex 2

2.

	Osaka	*Yokohama*
Average operating assets (a)	$1,000,000	$4,000,000
Net operating income.....	$210,000	$720,000
Minimum required return on average operating assets: 15% × (a).....	150,000	600,000
Residual income	$ 60,000	$120,000

Ex 3

In each of the cases below, assume Division X has a product that can be sold to outside customers or to Division Y of the same company. The managers of the divisions are evaluated based on their divisional profits.

	Case	
	A	B
Division X:		
Capacity in units	200,000	200,000
Number of units being sold to outside customers	200,000	160,000
Selling price per unit to outside customers	$90	$75
Variable costs per unit	$70	$60
Fixed costs per unit (based on capacity)	$13	$8
Division Y:		
Number of units needed for production	40,000	40,000
Purchase price per unit now being paid to an outside supplier	$86	$74

Required:

1. Refer to the data in case A above. Assume in this case $3 per unit in variable selling costs can be avoided on intracompany sales.
   a. What is the lowest acceptable transfer price from the perspective of the selling division?
   b. What is the highest acceptable transfer price from the perspective of the buying division?
   c. What is the range of acceptable transfer prices (if any) between the two divisions? If the managers are free to negotiate and make decisions on their own, will a transfer probably take place? Explain.
2. Refer to the data in case B above. In this case, there will be no savings in variable selling costs on intracompany sales.
   a. What is the lowest acceptable transfer price from the perspective of the selling division?
   b. What is the highest acceptable transfer price from the perspective of the buying division?
   c. What is the range of acceptable transfer prices (if any) between the two divisions? If the managers are free to negotiate and make decisions on their own, will a transfer probably take place? Explain.

Ex 3

1a. The lowest acceptable transfer price from the perspective of the selling division is given by the following formula:

$$\text{Transfer price} \geq \begin{matrix}\text{Variable cost} \\ \text{per unit}\end{matrix} + \frac{\begin{matrix}\text{Total contribution margin} \\ \text{on lost sales}\end{matrix}}{\text{Number of units transferred}}$$

There is no idle capacity, so each of the 40,000 units transferred from Division X to Division Y reduces sales to outsiders by one unit. The contribution margin per unit on outside sales is $20 (= $90 – $70).

$$\text{Transfer price} \geq (\$70 - \$3) + \frac{\$20 \times 40{,}000}{40{,}000}$$

$$\geq \$67 + \$20 = \$87$$

Ex 3

1b. The buying division, Division Y, can buy a similar unit from an outside supplier for $86. Therefore, Division Y would be unwilling to pay more than $86 per unit.

$$\text{Transfer price} \geq \text{Cost of buying from outside supplier} = \$86$$

1c. There is no range of acceptable transfer prices. The requirements of the two divisions are incompatible and no transfer will take place.

Ex 3

2a. In this case, Division X has enough idle capacity to satisfy Division Y's demand. Therefore, there are no lost sales and the lowest acceptable price as far as the selling division is concerned is the variable cost of $60 per unit.

$$\text{Transfer price} \geq \$60 + \frac{\$0}{40{,}000} = \$60$$

Ex 3

2b. The buying division, Division Y, can buy a similar unit from an outside supplier for $74. Therefore, Division Y would be unwilling to pay more than $74 per unit.

Transfer price £ Cost of buying from outside supplier = $74

Ex 3

2c. As shown below, the range of acceptable transfer prices is $60 to $74. In this case, the requirements of the two divisions are compatible and a transfer hopefully will take place.